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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore–607001
B. Sc Degree Examination - APRIL 2021

CH509 – ORGANIC CHEMISTRY - III

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Aniline on treatment with $NaNO_2$ and Hcl gives
   a) Benzene diazonium chloride b) Chlorobenzene
   c) Nitro benzene d) Anilinium chloride

2. Schiff's base on reduction give
   a) Aniline b) Primary amine c) Secondary amine d) Tertiary amine

3. The number of chiral centres in Tartaric acid is
   a) 1 b) 2 c) 3 d) 4

4. How many chiral stereoisomers can be drawn for $CH_3CHFCHFCH(CH_3)_2$
   a) 1 b) 3 c) 2 d) 4

5. Malonic ester react with acetaldehyde gives
   a) Crotonic acid b) Cinnamic acid c) Succinic acid d) Laevulinic acid

6. Tautomerism involved in nitroalkanes is
   a) Keto-enol b) Amido-imidol c) Nitro – acinitro d) Aldol

7. In the Beckmann reaction, migrating group is chiral; select the true statement for it
   a) It will change its configuration while it migration
   b) It will retain its configuration while it migration
   c) It can change as well as it can retain its configuration
   d) There is nothing to do with configuration

8. In which of the following rearrangement amide is reduced to amine which has one carbon less than the starting material?
   a) Lossen rearrangement b) Beckmann rearrangement
   c) Diekmann rearrangement d) Hoffmann rearrangement

9. Glucose reduce Fehling's solution and gives
   a) Blue precipitate b) Red precipitate c) Yellow precipitate d) Colourprecipitate

10. Which of the following is not a disaccharide
    a) Amylase b) Cellobiose c) Lactose d) None of these

II. Fill up the blanks

11. Nitrobenzene when reduced with Sn and HClgives _____

12. Hexane and 3-methyl pentane are example of _____

13. Diazo methane converts a carboxylic acid into its _____

14. Conversion of aromatic ketoximes to acid amide with PCl_5 is called _____

15. Example of monosaccharide is _____

III. Match the following

16. Shape of amine	-	Building blocks of proteins
17. Boat and chair conformation	-	Primary amine
18. Active methylene group	-	Pyranose sugars
19. Hoffmann rearrangement	-	Acetoacetic ester
20. Amino acids are	-	Pyramids

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. How aliphatic nitro compounds are prepared?
22. Write the preparation of diazonium salt from Aniline.
23. Define optical isomerism.
24. What are the conditions for a compound to be optically active?
25. What is Walden inversion?
26. Give an example for Claisen rearrangement.
27. Write the resonance structures of diazomethane.
28. What is Amido-Imidol tautomerism?
29. Define cationotropic rearrangement.
30. What is fries rearrangement?
31. How glucose can be converted to fructose?
32. What is Zwitter ion?

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) How m-dinitrobenzene is prepared? **(3)**
    b) Explain the substituent effect on basicity of aromatic amines. **(4)**

34. a) Explain the isomerism involved in lactic acid. **(3)**
    b) Discuss the R and S notation for isomers with one asymmetric carbon atom. **(4)**

35. a) Explain the elements of symmetry. **(4)**
    b) Discuss the optical activity of biphenyls? **(3)**

36. a) How is succinic acid and adipic acid are prepared from cyanoacetic ester? **(4)**
    b) Write any 2 synthetic uses of diazoacetic ester. **(3)**

37. a) Explain the reaction and mechanism of pinacol – pinacolone rearrangement. **(5)**
    b) Write the curtius rearrangement. **(2)**

38. a) Explain the reaction and mechanism of Beckmann rearrangement. **(4)**
    b) Write the osazone formation from glucose . **(3)**

39. a) Explain the structural elucidation of sucrose **(5)**
    b) Discuss the strecker synthesis of amino acids. **(2)**

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